

1)

```
mysql> SELECT *
```

```
-> FROM Employee
```

```
-> WHERE dept_id = 102;
```

```
+-----+-----+-----+-----+
| EmpID | EmpName | Salary | dept_id |
+-----+-----+-----+-----+
| 2 | Bala | 55000 | 102 |
| 3 | Charan | 60000 | 102 |
+-----+-----+-----+-----+
```

2 rows in set (0.00 sec)

```
mysql> SELECT EmpName
```

```
-> FROM Employee;
```

```
+-----+
| EmpName |
+-----+
| Arun |
| Bala |
| Charan |
| Divya |
| Esha |
```

```
mysql> SELECT EmpName
```

```
-> FROM Employee
```

```
-> WHERE dept_id = 101
```

```
->
```

```
-> UNION
```

```
->
```

```
-> SELECT EmpName
```

```
-> FROM Employee
```

```
-> WHERE dept_id = 102;
```

```
+-----+
| EmpName |
```

```
+-----+
| Arun   |
| Esha   |
| Bala   |
| Charan |
```

```
+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT EmpName
-> FROM Employee
-> WHERE dept_id NOT IN (102);
```

```
+-----+
| EmpName |
+-----+
| Arun   |
| Esha   |
| Divya  |
```

```
+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM Employee
-> CROSS JOIN Department;
```

```
+-----+-----+-----+-----+-----+-----+
| EmpID | EmpName | Salary | dept_id | dept_id | dept_name |
+-----+-----+-----+-----+-----+-----+
| 5 | Esha | 50000 | 101 | 10 | Bioinformatics |
| 4 | Divya | 45000 | 103 | 10 | Bioinformatics |
| 3 | Charan | 60000 | 102 | 10 | Bioinformatics |
```

|  |   |        |       |     |     |                  |  |
|--|---|--------|-------|-----|-----|------------------|--|
|  | 2 | Bala   | 55000 | 102 | 10  | Bioinformatics   |  |
|  | 1 | Arun   | 40000 | 101 | 10  | Bioinformatics   |  |
|  | 5 | Esha   | 50000 | 101 | 20  | Computer Science |  |
|  | 4 | Divya  | 45000 | 103 | 20  | Computer Science |  |
|  | 3 | Charan | 60000 | 102 | 20  | Computer Science |  |
|  | 2 | Bala   | 55000 | 102 | 20  | Computer Science |  |
|  | 1 | Arun   | 40000 | 101 | 20  | Computer Science |  |
|  | 5 | Esha   | 50000 | 101 | 30  | Biotechnology    |  |
|  | 4 | Divya  | 45000 | 103 | 30  | Biotechnology    |  |
|  | 3 | Charan | 60000 | 102 | 30  | Biotechnology    |  |
|  | 2 | Bala   | 55000 | 102 | 30  | Biotechnology    |  |
|  | 1 | Arun   | 40000 | 101 | 30  | Biotechnology    |  |
|  | 5 | Esha   | 50000 | 101 | 101 | HR               |  |
|  | 4 | Divya  | 45000 | 103 | 101 | HR               |  |
|  | 3 | Charan | 60000 | 102 | 101 | HR               |  |
|  | 2 | Bala   | 55000 | 102 | 101 | HR               |  |
|  | 1 | Arun   | 40000 | 101 | 101 | HR               |  |
|  | 5 | Esha   | 50000 | 101 | 102 | IT               |  |
|  | 4 | Divya  | 45000 | 103 | 102 | IT               |  |
|  | 3 | Charan | 60000 | 102 | 102 | IT               |  |
|  | 2 | Bala   | 55000 | 102 | 102 | IT               |  |
|  | 1 | Arun   | 40000 | 101 | 102 | IT               |  |
|  | 5 | Esha   | 50000 | 101 | 103 | Finance          |  |
|  | 4 | Divya  | 45000 | 103 | 103 | Finance          |  |
|  | 3 | Charan | 60000 | 102 | 103 | Finance          |  |
|  | 2 | Bala   | 55000 | 102 | 103 | Finance          |  |
|  | 1 | Arun   | 40000 | 101 | 103 | Finance          |  |

```
+-----+-----+-----+-----+-----+-----+
```

30 rows in set (0.00 sec)

mysql> SELECT e.EmpID,

```

-> e.EmpName,
-> e.Salary,
-> d.dept_name
-> FROM Employee e
-> INNER JOIN Department d
-> ON e.dept_id = d.dept_id;

```

```

+-----+-----+-----+-----+
| EmpID | EmpName | Salary | dept_name |
+-----+-----+-----+-----+
| 1 | Arun | 40000 | HR |
| 2 | Bala | 55000 | IT |
| 3 | Charan | 60000 | IT |
| 4 | Divya | 45000 | Finance |
| 5 | Esha | 50000 | HR |
+-----+-----+-----+-----+

```

2)

```

mysql> CREATE TABLE Department_Q2 (
-> dept_id INT PRIMARY KEY,
-> dept_name VARCHAR(50)
-> );

```

Query OK, 0 rows affected (0.03 sec)

```

mysql> CREATE TABLE Student_Q2 (
-> student_id INT PRIMARY KEY,
-> name VARCHAR(50),
-> dept_id INT,
-> FOREIGN KEY (dept_id) REFERENCES Department_Q2(dept_id)
-> );

```

Query OK, 0 rows affected (0.05 sec)

```

mysql> INSERT INTO Department_Q2 VALUES

```

```
-> (101,'HR'),
-> (102,'IT'),
-> (103,'Finance');
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Student_Q2 VALUES
```

```
-> (1,'Arun',101),
-> (2,'Bala',102),
-> (3,'Charan',102),
-> (4,'Divya',103),
-> (5,'Esha',101);
```

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Department_Q2;
```

```
+-----+-----+
| dept_id | dept_name |
+-----+-----+
| 101 | HR |
| 102 | IT |
| 103 | Finance |
+-----+-----+
```

3 rows in set (0.00 sec)

```
mysql> SELECT * FROM Student_Q2;
```

```
+-----+-----+-----+
| student_id | name | dept_id |
+-----+-----+-----+
| 1 | Arun | 101 |
| 2 | Bala | 102 |
```

|  |   |        |  |     |  |
|--|---|--------|--|-----|--|
|  | 3 | Charan |  | 102 |  |
|  | 4 | Divya  |  | 103 |  |
|  | 5 | Esha   |  | 101 |  |

+-----+-----+-----+

3)

mysql> CREATE TABLE Sales\_Q3 (

-> SaleID INT PRIMARY KEY,

-> DeptID INT,

-> Amount INT

-> );

Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Sales\_Q3 VALUES

-> (1,101,5000),

-> (2,101,7000),

-> (3,102,8000),

-> (4,102,6000),

-> (5,103,4000),

-> (6,103,3000);

Query OK, 6 rows affected (0.01 sec)

Records: 6 Duplicates: 0 Warnings: 0

mysql> SELECT \* FROM Sales\_Q3;

+-----+-----+-----+

|  |        |  |        |  |        |  |
|--|--------|--|--------|--|--------|--|
|  | SaleID |  | DeptID |  | Amount |  |
|--|--------|--|--------|--|--------|--|

+-----+-----+-----+

|  |   |  |     |  |      |  |
|--|---|--|-----|--|------|--|
|  | 1 |  | 101 |  | 5000 |  |
|--|---|--|-----|--|------|--|

|  |   |  |     |  |      |  |
|--|---|--|-----|--|------|--|
|  | 2 |  | 101 |  | 7000 |  |
|--|---|--|-----|--|------|--|

|  |   |  |     |  |      |  |
|--|---|--|-----|--|------|--|
|  | 3 |  | 102 |  | 8000 |  |
|--|---|--|-----|--|------|--|

|  |   |  |     |  |      |  |
|--|---|--|-----|--|------|--|
|  | 4 |  | 102 |  | 6000 |  |
|--|---|--|-----|--|------|--|

|  |   |  |     |  |      |  |
|--|---|--|-----|--|------|--|
|  | 5 |  | 103 |  | 4000 |  |
|--|---|--|-----|--|------|--|

```
| 6 | 103 | 3000 |
```

```
+-----+-----+-----+
```

6 rows in set (0.00 sec)

```
mysql> SELECT DeptID,
```

```
-> COUNT(*) AS TotalSales,
```

```
-> SUM(Amount) AS TotalAmount,
```

```
-> AVG(Amount) AS AverageAmount,
```

```
-> MAX(Amount) AS MaximumSale,
```

```
-> MIN(Amount) AS MinimumSale
```

```
-> FROM Sales_Q3
```

```
-> GROUP BY DeptID
```

```
-> HAVING SUM(Amount) > 10000;
```

```
+-----+-----+-----+-----+-----+-----+
```

```
| DeptID | TotalSales | TotalAmount | AverageAmount | MaximumSale | MinimumSale |
```

```
+-----+-----+-----+-----+-----+-----+
```

```
| 101 | 2 | 12000 | 6000.0000 | 7000 | 5000 |
```

```
| 102 | 2 | 14000 | 7000.0000 | 8000 | 6000 |
```

```
+-----+-----+-----+-----+-----+-----+
```

2 rows in set (0.00 sec)

4)

```
mysql> SELECT EmpID,
```

```
-> EmpName,
```

```
-> Salary,
```

```
-> dept_id
```

```
-> FROM Employee e
```

```
-> WHERE Salary >
```

```
-> (
```

```
-> SELECT AVG(Salary)
```

```
-> FROM Employee
```

```
-> WHERE dept_id = e.dept_id
```

```
-> );
```

```

+-----+-----+-----+-----+
| EmpID | EmpName | Salary | dept_id |
+-----+-----+-----+-----+
| 3 | Charan | 60000 | 102 |
| 5 | Esha | 50000 | 101 |
+-----+-----+-----+-----+

```

5)

```

mysql> CREATE TABLE Project_Q5 (
-> ProjectID INT PRIMARY KEY,
-> ProjectName VARCHAR(50),
-> EmplID INT
-> );

```

Query OK, 0 rows affected (0.03 sec)

```

mysql> INSERT INTO Project_Q5 VALUES
-> (1,'Library System',1),
-> (2,'Banking App',2),
-> (3,'Hospital System',3),
-> (4,'E-Commerce',6);

```

Query OK, 4 rows affected (0.01 sec)

Records: 4 Duplicates: 0 Warnings: 0

```

mysql> SELECT e.EmpID,
-> e.EmpName,
-> p.ProjectName
-> FROM Employee e
-> INNER JOIN Project_Q5 p
-> ON e.EmpID = p.EmpID;

```

```

+-----+-----+-----+
| EmpID | EmpName | ProjectName |
+-----+-----+-----+

```

|   |        |                 |
|---|--------|-----------------|
| 1 | Arun   | Library System  |
| 2 | Bala   | Banking App     |
| 3 | Charan | Hospital System |

+-----+-----+-----+

3 rows in set (0.00 sec)

```
mysql> SELECT e.EmpID,
-> e.EmpName,
-> p.ProjectName
-> FROM Employee e
-> LEFT JOIN Project_Q5 p
-> ON e.EmpID = p.EmpID;
```

+-----+-----+-----+

| EmpID | EmpName | ProjectName |
|-------|---------|-------------|
|-------|---------|-------------|

+-----+-----+-----+

|   |        |                 |
|---|--------|-----------------|
| 1 | Arun   | Library System  |
| 2 | Bala   | Banking App     |
| 3 | Charan | Hospital System |
| 4 | Divya  | NULL            |
| 5 | Esha   | NULL            |

+-----+-----+-----+

5 rows in set (0.00 sec)

```
mysql> SELECT e.EmpID,
-> e.EmpName,
-> p.ProjectName
-> FROM Employee e
-> RIGHT JOIN Project_Q5 p
-> ON e.EmpID = p.EmpID;
```

+-----+-----+-----+

| EmpID | EmpName | ProjectName |
|-------|---------|-------------|
|-------|---------|-------------|

+-----+-----+-----+

|   |      |                |
|---|------|----------------|
| 1 | Arun | Library System |
|---|------|----------------|

|      |        |                 |
|------|--------|-----------------|
| 2    | Bala   | Banking App     |
| 3    | Charan | Hospital System |
| NULL | NULL   | E-Commerce      |

+-----+-----+-----+

4 rows in set (0.00 sec)

mysql> SELECT e.EmpID,

-> e.EmpName,

-> p.ProjectName

-> FROM Employee e

-> LEFT JOIN Project\_Q5 p

-> ON e.EmpID = p.EmpID

->

-> UNION

->

-> SELECT e.EmpID,

-> e.EmpName,

-> p.ProjectName

-> FROM Employee e

-> RIGHT JOIN Project\_Q5 p

-> ON e.EmpID = p.EmpID;

+-----+-----+-----+

|       |         |             |
|-------|---------|-------------|
| EmpID | EmpName | ProjectName |
|-------|---------|-------------|

+-----+-----+-----+

|      |        |                 |
|------|--------|-----------------|
| 1    | Arun   | Library System  |
| 2    | Bala   | Banking App     |
| 3    | Charan | Hospital System |
| 4    | Divya  | NULL            |
| 5    | Esha   | NULL            |
| NULL | NULL   | E-Commerce      |

+-----+-----+-----+

6 rows in set (0.00 sec)

```
mysql> SELECT e.EmpName,
->    p.ProjectName
-> FROM Employee e
-> CROSS JOIN Project_Q5 p;
```

```
+-----+-----+
| EmpName | ProjectName |
+-----+-----+
| Arun    | E-Commerce  |
| Arun    | Hospital System |
| Arun    | Banking App  |
| Arun    | Library System |
| Bala    | E-Commerce  |
| Bala    | Hospital System |
| Bala    | Banking App  |
| Bala    | Library System |
| Charan  | E-Commerce  |
| Charan  | Hospital System |
| Charan  | Banking App  |
| Charan  | Library System |
| Divya   | E-Commerce  |
| Divya   | Hospital System |
| Divya   | Banking App  |
| Divya   | Library System |
| Esha    | E-Commerce  |
| Esha    | Hospital System |
| Esha    | Banking App  |
| Esha    | Library System |
+-----+-----+
```

6)

```
mysql> CREATE VIEW DeptSalaryView AS
-> SELECT d.dept_name,
```

```

-> COUNT(e.EmpID) AS TotalEmployees,
-> AVG(e.Salary) AS AverageSalary
-> FROM Department d
-> JOIN Employee e
-> ON d.dept_id = e.dept_id
-> GROUP BY d.dept_name;

```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM DeptSalaryView;
```

```

+-----+-----+-----+
| dept_name | TotalEmployees | AverageSalary |
+-----+-----+-----+
| HR      | 2 | 45000.0000 |
| IT      | 2 | 57500.0000 |
| Finance | 1 | 45000.0000 |
+-----+-----+-----+

```

7)

```
mysql> CREATE TABLE Course_Q7 (
```

```

-> CourseID INT PRIMARY KEY,
-> CourseName VARCHAR(50)
-> );

```

Query OK, 0 rows affected (0.02 sec)

```
mysql> CREATE TABLE Enrollment_Q7 (
```

```

-> StudentID INT,
-> CourseID INT
-> );

```

Query OK, 0 rows affected (0.02 sec)

```
mysql> INSERT INTO Course_Q7 VALUES
```

```
-> (1,'DBMS'),  
-> (2,'Java'),  
-> (3,'Python');
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Enrollment_Q7 VALUES
```

```
-> (101,1),  
-> (101,2),  
-> (101,3),  
-> (102,1),  
-> (102,2),  
-> (103,1),  
-> (103,2),  
-> (103,3);
```

Query OK, 8 rows affected (0.01 sec)

Records: 8 Duplicates: 0 Warnings: 0

```
mysql> SELECT StudentID
```

```
-> FROM Enrollment_Q7  
-> GROUP BY StudentID  
-> HAVING COUNT(DISTINCT CourseID) =  
-> (  
-> SELECT COUNT(*)  
-> FROM Course_Q7  
-> );
```

```
+-----+
```

```
| StudentID |
```

```
+-----+
```

```
| 101 |
```

```
| 103 |
```

```
+-----+
```

2 rows in set (0.01 sec)

8)

```
mysql> SELECT * FROM Student_Q2;
```

```
+-----+-----+-----+
```

```
| student_id | name  | dept_id |
```

```
+-----+-----+-----+
```

```
|      1 | Arun  |   101 |
```

```
|      2 | Bala  |   102 |
```

```
|      3 | Charan |   102 |
```

```
|      4 | Divya |   103 |
```

```
|      5 | Esha  |   101 |
```

```
+-----+-----+-----+
```

5 rows in set (0.00 sec)

```
mysql> INSERT INTO Student_Q2
```

```
  -> VALUES (6,'Karthik',102);
```

Query OK, 1 row affected (0.01 sec)

```
mysql> SELECT * FROM Student_Q2;
```

```
+-----+-----+-----+
```

```
| student_id | name  | dept_id |
```

```
+-----+-----+-----+
```

```
|      1 | Arun  |   101 |
```

```
|      2 | Bala  |   102 |
```

```
|      3 | Charan |   102 |
```

```
|      4 | Divya |   103 |
```

```
|      5 | Esha  |   101 |
```

```
|      6 | Karthik |   102 |
```

```
+-----+-----+-----+
```

6 rows in set (0.00 sec)

```
mysql> UPDATE Student_Q2
```

```
-> SET name = 'Karthik Kumar'
```

```
-> WHERE student_id = 6;
```

```
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM Student_Q2
```

```
-> WHERE student_id = 6;
```

```
+-----+-----+-----+
| student_id | name      | dept_id |
+-----+-----+-----+
|      6 | Karthik Kumar |    102 |
+-----+-----+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> ^C
```

```
mysql> DELETE FROM Student_Q2
```

```
-> WHERE student_id = 6;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM Student_Q2;
```

```
+-----+-----+-----+
| student_id | name      | dept_id |
+-----+-----+-----+
|      1 | Arun      |    101 |
|      2 | Bala      |    102 |
|      3 | Charan    |    102 |
|      4 | Divya     |    103 |
|      5 | Esha      |    101 |
+-----+-----+-----+
```

5 rows in set (0.00 sec)

9)

```
mysql> SELECT *
```

```
-> FROM Employee
```

```
-> WHERE dept_id = 102
```

```
-> AND Salary > 50000;
```

```
+-----+-----+-----+-----+
| EmpID | EmpName | Salary | dept_id |
+-----+-----+-----+-----+
|  2   | Bala   | 55000  |    102  |
|  3   | Charan | 60000  |    102  |
+-----+-----+-----+-----+
```

2 rows in set (0.00 sec)

10)

```
mysql> CREATE TABLE Passenger_Q10 (
```

```
-> PassengerID INT PRIMARY KEY,
```

```
-> Name VARCHAR(50),
```

```
-> Phone VARCHAR(15)
```

```
-> );
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> CREATE TABLE Flight_Q10 (
```

```
-> FlightID INT PRIMARY KEY,
```

```
-> FlightName VARCHAR(50),
```

```
-> Source VARCHAR(30),
```

```
-> Destination VARCHAR(30)
```

```
-> );
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> CREATE TABLE Booking_Q10 (
```

```
-> BookingID INT PRIMARY KEY,  
-> PassengerID INT,  
-> FlightID INT,  
-> JourneyDate DATE,  
-> FOREIGN KEY (PassengerID) REFERENCES Passenger_Q10(PassengerID),  
-> FOREIGN KEY (FlightID) REFERENCES Flight_Q10(FlightID)  
-> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> INSERT INTO Passenger_Q10 VALUES
```

```
-> (1,'Arun','9876543210'),  
-> (2,'Bala','9876543211'),  
-> (3,'Charan','9876543212');
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Flight_Q10 VALUES
```

```
-> (101,'IndiGo','Chennai','Delhi'),  
-> (102,'Air India','Chennai','Mumbai'),  
-> (103,'SpiceJet','Chennai','Bangalore');
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Booking_Q10 VALUES
```

```
-> (1001,1,101,'2026-08-10'),  
-> (1002,2,102,'2026-08-12'),  
-> (1003,3,103,'2026-08-15');
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Passenger_Q10;
```

```

+-----+-----+-----+
| PassengerID | Name | Phone |
+-----+-----+-----+
| 1 | Arun | 9876543210 |
| 2 | Bala | 9876543211 |
| 3 | Charan | 9876543212 |
+-----+-----+-----+

```

3 rows in set (0.00 sec)

mysql> SELECT \* FROM Flight\_Q10;

```

+-----+-----+-----+-----+
| FlightID | FlightName | Source | Destination |
+-----+-----+-----+-----+
| 101 | IndiGo | Chennai | Delhi |
| 102 | Air India | Chennai | Mumbai |
| 103 | SpiceJet | Chennai | Bangalore |
+-----+-----+-----+-----+

```

3 rows in set (0.00 sec)

mysql> SELECT \* FROM Booking\_Q10;

```

+-----+-----+-----+-----+
| BookingID | PassengerID | FlightID | JourneyDate |
+-----+-----+-----+-----+
| 1001 | 1 | 101 | 2026-08-10 |
| 1002 | 2 | 102 | 2026-08-12 |
| 1003 | 3 | 103 | 2026-08-15 |
+-----+-----+-----+-----+

```

3 rows in set (0.00 sec)

mysql> SELECT p.Name,

-> f.FlightName

```

-> FROM Passenger_Q10 p
-> JOIN Booking_Q10 b
-> ON p.PassengerID = b.PassengerID
-> JOIN Flight_Q10 f
-> ON b.FlightID = f.FlightID;

```

```

+-----+-----+
| Name  | FlightName |
+-----+-----+
| Arun  | IndiGo    |
| Bala  | Air India |
| Charan | SpiceJet  |
+-----+-----+

```

3 rows in set (0.00 sec)

mysql> SELECT Name

```

-> FROM Passenger_Q10
-> WHERE PassengerID IN
-> (
->   SELECT PassengerID
->   FROM Booking_Q10
-> );

```

```

+-----+
| Name  |
+-----+
| Arun  |
| Bala  |
| Charan |
+-----+

```

3 rows in set (0.00 sec)

mysql> SELECT FlightID,

-> COUNT(\*) AS TotalBookings

-> FROM Booking\_Q10

-> GROUP BY FlightID;

```
+-----+-----+
| FlightID | TotalBookings |
+-----+-----+
| 101 | 1 |
| 102 | 1 |
| 103 | 1 |
+-----+-----+
```

3 rows in set (0.00 sec)

mysql> CREATE VIEW BookingDetails\_Q10 AS

-> SELECT p.Name,

-> f.FlightName,

-> b.JourneyDate

-> FROM Passenger\_Q10 p

-> JOIN Booking\_Q10 b

-> ON p.PassengerID = b.PassengerID

-> JOIN Flight\_Q10 f

-> ON b.FlightID = f.FlightID;

Query OK, 0 rows affected (0.01 sec)

mysql> SELECT \* FROM BookingDetails\_Q10;

```
+-----+-----+-----+
| Name | FlightName | JourneyDate |
+-----+-----+-----+
| Arun | IndiGo | 2026-08-10 |
| Bala | Air India | 2026-08-12 |
| Charan | SpiceJet | 2026-08-15 |
+-----+-----+-----+
```

3 rows in set (0.00 sec)